

Unit 1 Question: Introduction To Information Security

Mark:4

Q 1: A student receives a phishing email asking for login credentials. Explain the steps the student should follow to avoid becoming a victim.

When a student receives a phishing email asking for login credentials, the following steps should be followed to avoid becoming a victim:

1. **Do not click on any links or attachments** in the email, as they may lead to fake websites or install malware.
2. **Verify the sender's email address** and check for spelling mistakes, suspicious domains, or unusual requests.
3. **Never share login credentials or personal information** through email, even if the message appears to come from a trusted organization.
4. **Report the phishing email** to the institution's IT department or email administrator and delete the email immediately.

Result: By following these steps, the student can protect their account, personal information, and device from phishing attacks.

Q:2 Apply the characteristics of a strong password to create a password suitable for an online banking account.

A strong password for an online banking account should have the following characteristics:

1. **Use at least 12–16 characters** to increase security.
2. **Include a mix of uppercase letters, lowercase letters, numbers, and special characters.**
3. **Avoid personal information** such as names, birthdays, phone numbers, or common words.
4. **Use a unique password** that is not reused for any other account.

Example of a strong password:

R8!mQ#27vLp@9Xz

Note: In addition to using a strong password, enable **Multi-Factor Authentication (MFA)** for better protection of online banking accounts.

Q3: A user connects to public Wi-Fi at an airport. Explain how VPN helps protect the user's data.

When a user connects to public Wi-Fi at an airport, a **Virtual Private Network (VPN)** helps protect the user's data in the following ways:

1. **Encrypts Internet Traffic:** VPN encrypts all data sent between the user's device and the VPN server, making it unreadable to attackers.
2. **Protects Sensitive Information:** Login credentials, banking details, and personal information remain secure even on unsecured public Wi-Fi.
3. **Prevents Eavesdropping:** Cybercriminals on the same Wi-Fi network cannot easily intercept or monitor the user's online activities.
4. **Masks the User's IP Address:** VPN hides the user's real IP address, improving privacy and making online tracking more difficult.

Result: Using a VPN on public Wi-Fi significantly reduces the risk of data theft, hacking, and unauthorized access.

Q4: A shopping website collects customer information. Apply any four privacy principles that should be followed while handling customer data.

A shopping website should follow these privacy principles while handling customer data:

1. **Collect Only Necessary Data:** Gather only the information required to process orders and provide services.
2. **Obtain User Consent:** Inform customers about data collection and obtain their permission before collecting or using personal information.
3. **Protect Customer Data:** Store customer information securely using encryption, access control, and regular security updates.
4. **Use Data Only for the Intended Purpose:** Customer data should be used only for order processing and related services, and should not be shared with third parties without permission.

Result: Following these privacy principles helps protect customer information, builds user trust, and ensures responsible handling of personal data.

Q5: Compare Malware and Identity Theft with suitable examples.

Aspect	Malware	Identity Theft
Definition	Malicious software designed to damage, disrupt, or gain unauthorized access to a computer system.	A crime in which someone steals and misuses another person's personal information.
Target	Computers, mobile devices, and networks.	Personal identity, financial, and confidential information.
Purpose	To steal data, damage files, or control the victim's device.	To commit fraud, access bank accounts, or impersonate the victim.
Example	Ransomware encrypts files and demands payment to restore access.	A cybercriminal uses a stolen Aadhaar number and bank credentials to make unauthorized transactions.

Conclusion: Malware attacks computer systems using malicious software, whereas identity theft targets a person's personal information for fraudulent purposes.

Q6: Analyze the differences between Hardware Firewall and Software Firewall.

Aspect	Hardware Firewall	Software Firewall
Installation	Installed as a dedicated physical device between the network and the Internet.	Installed as an application on an individual computer or server.
Protection	Protects the entire network from external threats.	Protects only the device on which it is installed.
Performance	Does not consume the host system's CPU or memory.	Uses the computer's CPU, memory, and storage resources.
Example	Cisco ASA, Fortinet FortiGate, Sophos XGS Firewall.	Windows Defender Firewall, UFW (Linux), ZoneAlarm.

Conclusion: A hardware firewall is suitable for securing an entire network, while a software firewall is ideal for protecting individual devices.

Q7: Compare Administrative, Technical and Physical Security Controls.

Aspect	Administrative Controls	Technical Controls	Physical Controls
Purpose	Establish security policies, procedures, and user responsibilities.	Protect systems and data using technology.	Prevent unauthorized physical access to facilities and equipment.
Implementation	Security policies, employee training, password policies, incident response plans.	Firewalls, antivirus software, encryption, Multi-Factor Authentication (MFA).	CCTV cameras, biometric access, security guards, door locks.
Focus	Managing people and processes.	Securing digital systems and networks.	Securing buildings, rooms, and hardware.
Example	Password policy and security awareness training.	Firewall blocking unauthorized traffic.	Fingerprint scanner controlling entry to the server room.

Conclusion: Administrative controls manage people and policies, technical controls protect digital assets using technology, and physical controls secure the organization's physical infrastructure.

Q8: Analyze the differences between Copyright and Patent with suitable examples.

Aspect	Copyright	Patent
Purpose	Protects original creative works such as books, music, software, and artwork.	Protects new inventions, products, processes, or technological innovations.
What is Protected	Expression of ideas (e.g., source code, books, videos, music).	Functional inventions and innovative technical solutions.
Rights Given	Gives the creator the exclusive right to reproduce, distribute, and publish the work.	Gives the inventor the exclusive right to make, use, sell, or license the invention.
Example	Copyright protection for a software program, novel, or movie.	Patent for a new medical device, smartphone technology, or innovative networking algorithm.

Conclusion: Copyright protects **creative works**, whereas a patent protects **new inventions and technological innovations**.

Q9: Apply the Security Management Cycle to improve the security of a college computer laboratory. 6 Marks Apply (L3)

The **Security Management Cycle** can be applied to improve the security of a college computer laboratory through the following steps:

1. Identify Assets and Risks

- Identify important assets such as computers, servers, student data, software, and network resources.
- Analyze possible threats like malware attacks, unauthorized access, and data theft.

2. Develop Security Policies

- Create rules for password usage, internet access, software installation, and acceptable computer usage.
- Define user access permissions for students, faculty, and administrators.

3. Implement Security Controls

- Install antivirus software, firewalls, and regular security updates.
- Enable user authentication, access control, and data backup mechanisms.

4. Monitor Security Activities

- Regularly monitor network traffic, login activities, and system logs.
- Check for suspicious activities or unauthorized access attempts.

5. Respond to Security Incidents

- Take immediate action during incidents such as malware infection or unauthorized access.
- Isolate affected systems and restore data from backups.

6. Review and Improve Security

- Conduct periodic security audits and update security policies.
- Provide cybersecurity awareness training to students and staff.

Conclusion: Applying the Security Management Cycle helps the college laboratory continuously identify risks, implement protection measures, and improve overall security.

Q10: Explain how HTTPS and SSL/TLS together provide secure web communication during online banking. 6 Apply (L3)

During online banking, **HTTPS and SSL/TLS work together to provide secure communication between the user's browser and the banking server** through the following steps:

1. Establishing a Secure Connection

- When a user accesses a banking website using **HTTPS**, the browser starts an SSL/TLS handshake with the bank's server.
- The server provides a digital certificate to prove its identity.

2. Authentication Using Digital Certificates

- SSL/TLS verifies the bank's identity using a certificate issued by a trusted Certificate Authority (CA).
- This prevents users from connecting to fake or phishing websites.

3. Data Encryption

- SSL/TLS encrypts data exchanged between the browser and server.
- Sensitive information such as usernames, passwords, and transaction details cannot be read by attackers.

4. Data Integrity Protection

- SSL/TLS uses cryptographic methods to ensure that data is not modified during transmission.
- Any unauthorized changes in data are detected.

5. Secure Data Transfer through HTTPS

- HTTPS is the combination of HTTP with SSL/TLS security features.
- It ensures that all communication between the customer and bank remains confidential and secure.

6. Protection Against Cyber Attacks

- HTTPS with SSL/TLS helps prevent attacks such as eavesdropping, man-in-the-middle attacks, and data theft.

Conclusion: HTTPS provides secure web communication by using SSL/TLS encryption, authentication, and integrity mechanisms, making online banking transactions safer for users.

Q11: A company stores important documents digitally. Explain how encryption and regular backup together protect organizational data.
6 Apply (L3)

A company can protect its digital documents by using **encryption and regular backup together** as follows:

1. Encryption of Data

- Encryption converts important documents into an unreadable format using cryptographic algorithms.
- Only authorized users with the correct encryption key can access the original data.
- It protects confidential information from unauthorized access and data theft.

2. Protection During Data Transfer and Storage

- Encrypted documents remain secure while stored on servers, cloud storage, or during transmission over networks.
- Even if attackers gain access to the storage system, they cannot easily read the encrypted data.

3. Regular Data Backup

- Regular backups create additional copies of important documents on separate storage locations.
- Backups help recover data in case of accidental deletion, hardware failure, malware attacks, or ransomware incidents.

4. Disaster Recovery

- If the original data is damaged or lost, the organization can restore files from backup copies.
- This reduces downtime and ensures business continuity.

5. Combining Encryption and Backup

- Backup files should also be encrypted to prevent unauthorized access.
- A combination of encrypted storage and secure backups provides both confidentiality and availability of data.

6. Example

- A company encrypts its financial reports before storing them on a cloud server and maintains daily encrypted backups. If ransomware attacks the system, the company can restore the reports safely from backups.

Conclusion: Encryption protects data from unauthorized access, while regular backups ensure data recovery. Together, they provide strong protection against data loss and security threats.

Q12: Compare Packet Filtering Firewall and Stateful Inspection Firewall based on working, advantages and limitations. 6 Analyze (L4)

Aspect	Packet Filtering Firewall	Stateful Inspection Firewall
Working	Examines each packet independently based on source/destination IP address, port number, and protocol.	Monitors packets and keeps track of the state of active connections before allowing traffic.
Advantages	Fast, simple to configure, and uses fewer system resources.	Provides better security by allowing only packets that belong to valid, established sessions.
Limitations	Cannot identify the context of a connection and is vulnerable to IP spoofing and certain attacks.	Requires more memory and processing power and is more complex to configure than packet filtering.
Security Level	Basic security with limited traffic analysis.	Higher security through connection-aware inspection.
Performance	High performance due to simple rule checking.	Slightly slower because it maintains a state table for active connections.
Example	Basic router Access Control Lists (ACLs).	Cisco ASA Stateful Firewall, Check Point Stateful Firewall.

Conclusion

A **Packet Filtering Firewall** is suitable for basic and high-speed traffic filtering, whereas a **Stateful Inspection Firewall** provides stronger security by tracking active connections and blocking unauthorized or suspicious traffic.

Q13: Analyze the relationship among Assets, Threats, Vulnerabilities, Risks and Controls using a suitable example. 6 Analyze (L4)

Component	Description	Example (College Student Database)
Asset	Anything valuable that needs protection.	Student database containing personal and academic records.
Threat	A potential event that can harm an asset.	A hacker attempts to steal student information.
Vulnerability	A weakness that can be exploited by a threat.	Weak passwords and outdated server software.
Risk	The possibility of loss when a threat exploits a vulnerability.	Unauthorized access leading to data theft or modification.
Control	A safeguard implemented to reduce or eliminate risk.	Strong passwords, Multi-Factor Authentication (MFA), firewall, encryption, and regular software updates.

Example Scenario

A college stores student records (**Asset**) on a server. A cybercriminal (**Threat**) exploits weak passwords (**Vulnerability**) to gain unauthorized access, resulting in data theft (**Risk**). The college implements **Controls** such as strong passwords, MFA, encryption, firewalls, and regular security updates to protect the database.

Conclusion: Assets are protected by implementing appropriate **security controls** that reduce the **risk** of **threats** exploiting **vulnerabilities**.

Q14: Analyze the working of asymmetric cryptography using Alice and Bob communication. 6 Analyze (L4)

Asymmetric cryptography uses a **pair of keys**: a **public key** for encryption and a **private key** for decryption. The public key can be shared with everyone, while the private key is kept secret.

Working of Asymmetric Cryptography (Alice and Bob)

1. Key Generation

- Bob generates two keys:
 - **Public Key** (shared with everyone)
 - **Private Key** (kept secret)

2. Public Key Sharing

- Bob sends his **public key** to Alice.
- Bob never shares his **private key**.

3. Message Encryption

- Alice writes a confidential message.
- She encrypts the message using **Bob's public key**.

4. Secure Transmission

- The encrypted message is sent over the Internet.
- Even if an attacker intercepts it, the message cannot be read.

5. Message Decryption

- Bob receives the encrypted message.
- He decrypts it using **his private key**.

6. Secure Communication

- Since only Bob possesses the private key, only he can read the original message, ensuring confidentiality.

Example:

- **Alice's Message:** "Transfer ₹10,000."
- Alice encrypts the message using **Bob's public key**.
- Bob decrypts it using **his private key** and reads the original message.

Conclusion

Asymmetric cryptography provides **secure communication** by using a **public key for encryption** and a **private key for decryption**, ensuring that only the intended recipient (Bob) can access the original message.

Q15: Evaluate the effectiveness of software updates in preventing cyber attacks. 6 Evaluate (L5)

Software updates are one of the most effective methods for preventing cyber attacks because they fix security weaknesses and improve system protection.

Evaluation of Effectiveness

1. **Fixes Security Vulnerabilities**
 - Updates patch known software flaws that attackers can exploit.
 - This reduces the chances of hacking and unauthorized access.
2. **Protects Against New Threats**
 - Regular updates include protection against newly discovered malware, ransomware, and phishing techniques.
 - Systems remain secure against evolving cyber threats.
3. **Improves System Stability and Performance**
 - Updates not only improve security but also fix bugs and enhance the reliability of the operating system and applications.
4. **Enhances Security Features**
 - New versions often introduce stronger encryption, better authentication methods, and improved access controls.
5. **Limitations**
 - Updates alone cannot prevent all cyber attacks.
 - Users must also use strong passwords, enable Multi-Factor Authentication (MFA), install antivirus software, and practice safe browsing habits.
6. **Overall Evaluation**
 - Regular software updates significantly reduce cyber risks by closing security gaps.
 - When combined with other security measures, they provide strong protection against cyber attacks.

Conclusion: Software updates are highly effective in preventing cyber attacks because they remove known vulnerabilities and strengthen system security. However, they should be combined with other cybersecurity practices for comprehensive protection.

Q16: Evaluate the importance of Digital Certificates in secure Internet communication. 6 Evaluate (L5)

Digital Certificates are electronic documents issued by a **Certificate Authority (CA)** that verify the identity of a website, server, or organization. They play a vital role in securing Internet communication.

Evaluation of Importance

1. **Authenticates Identity**
 - Digital certificates verify that a website or server is genuine.
 - This helps users avoid fake or phishing websites.
2. **Enables Secure Communication**
 - They are used in **SSL/TLS** to establish secure **HTTPS** connections.
 - Data exchanged between the client and server is encrypted.
3. **Ensures Data Confidentiality**
 - Information such as passwords, banking details, and personal data remains protected from unauthorized access during transmission.
4. **Maintains Data Integrity**
 - Digital certificates help ensure that transmitted data is not altered while traveling across the network.
 - Any tampering can be detected.
5. **Builds User Trust**
 - Browsers display a padlock icon for websites with valid certificates.
 - Users are more confident when sharing sensitive information on trusted websites.
6. **Overall Evaluation**
 - Without digital certificates, users cannot easily verify website authenticity, increasing the risk of phishing and man-in-the-middle attacks.
 - Therefore, digital certificates are essential for secure and trusted online communication.

Conclusion: Digital certificates are crucial for Internet security because they provide **authentication, encryption, data integrity, and user trust**, making online services such as banking, e-commerce, and email communication secure.

Q17: Evaluate the consequences of software piracy for individuals and organizations. 6 Evaluate (L5)

Software piracy is the unauthorized copying, installation, distribution, or use of copyrighted software without a valid license. It has serious consequences for both individuals and organizations.

Evaluation of Consequences

1. Legal Consequences

- Individuals and organizations may face legal action, heavy fines, or penalties for violating copyright laws.

2. Security Risks

- Pirated software may contain malware, viruses, ransomware, or spyware, leading to data theft and system compromise.

3. No Updates or Technical Support

- Users of pirated software do not receive official security updates, bug fixes, or technical support, making systems more vulnerable to cyber attacks.

4. Financial Loss

- Organizations may suffer financial losses due to legal penalties, recovery costs after malware infections, and reduced employee productivity.

5. Reputation Damage

- Using illegal software can damage the reputation of an organization and reduce customer trust.

6. Overall Evaluation

- Although pirated software may appear to reduce initial costs, the legal, security, financial, and reputational risks far outweigh any short-term savings.

Conclusion: Software piracy has severe consequences, including **legal penalties, security threats, financial losses, lack of support, and reputational damage**. Therefore, individuals and organizations should always use genuine, licensed software.

Q18: Compare Proxy Firewall and Next-Generation Firewall (NGFW) with respect to architecture, features, advantages, disadvantages and applications. 8 marks Analyze (L4)

Aspect	Proxy Firewall	Next-Generation Firewall (NGFW)
Architecture	Acts as an intermediary between the client and the destination server. It receives, inspects, and forwards requests.	Combines traditional firewall functions with advanced security features such as Deep Packet Inspection (DPI), Intrusion Prevention System (IPS), and application awareness.
Features	Hides internal IP addresses, filters web traffic, performs content filtering, and supports user authentication.	Stateful inspection, Deep Packet Inspection (DPI), application control, IPS, antivirus, malware protection, SSL/TLS inspection, and user identity awareness.
Advantages	Provides strong privacy by hiding internal systems, filters application-layer traffic, and improves security through proxy-based communication.	Offers comprehensive protection against modern cyber threats, detects application-layer attacks, controls applications, and provides high network visibility.
Disadvantages	May introduce higher latency, consumes more resources, and supports limited modern threat detection.	More expensive, requires skilled administrators, and needs greater processing power.
Applications	Used in web proxies, email gateways, content filtering, and organizations requiring controlled Internet access.	Used in enterprise networks, data centers, banks, universities, cloud environments, and organizations requiring advanced cybersecurity.

Conclusion

- **Proxy Firewall** is suitable for organizations that need **secure web access, privacy, and content filtering**.
- **Next-Generation Firewall (NGFW)** is ideal for **modern enterprise networks** because it provides **advanced threat detection, application awareness, intrusion prevention, and comprehensive network security** against sophisticated cyber attacks.

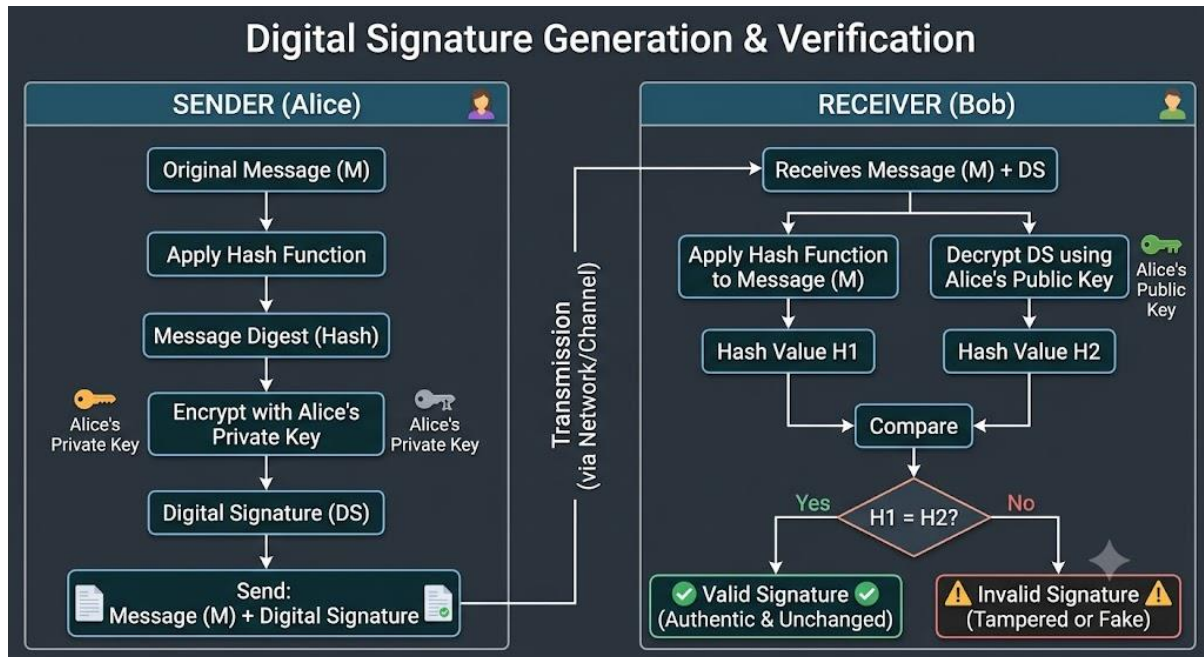
Q19: Analyze the role of Internet Security protocols (HTTPS, SSL/TLS, VPN and Secure Email) in protecting Internet communication. 8
Analyze (L4)

Protocol	Role in Internet Security	Key Features	Example Application
HTTPS	Secures communication between a web browser and a web server.	Encrypts web traffic, authenticates websites, ensures data integrity.	Online banking, e-commerce, and online payment websites.
SSL/TLS	Provides encryption and authentication for Internet communication. It is the security protocol used by HTTPS.	Data encryption, digital certificates, authentication, integrity protection.	Secure websites, email services, and online transactions.
VPN (Virtual Private Network)	Creates an encrypted tunnel between the user's device and a VPN server over the Internet.	Encrypts all Internet traffic, hides IP address, protects data on public Wi-Fi.	Remote employee access and secure browsing on public Wi-Fi.
Secure Email	Protects email communication from unauthorized access and tampering.	Email encryption, digital signatures, sender authentication, message integrity.	Secure exchange of confidential business and financial information.

Conclusion

HTTPS, SSL/TLS, VPN, and Secure Email each protect different aspects of Internet communication. Together, they provide **confidentiality, authentication, integrity, and privacy**, ensuring secure web browsing, remote access, and email communication while reducing the risk of cyber attacks and data theft.

Q20: Analyze the complete process of Digital Signature generation and verification with a neat diagram. 8 Analyze (L4)



Working of Digital Signature

1. **Message Creation**
 - Alice prepares the original message (M).
2. **Hash Generation**
 - A hash function (e.g., SHA-256) generates a fixed-size **message digest** from the original message.
3. **Digital Signature Generation**
 - Alice encrypts the message digest using **her private key**.
 - The encrypted digest becomes the **Digital Signature**.
4. **Message Transmission**
 - Alice sends both the **original message** and the **digital signature** to Bob.
5. **Verification at Receiver**
 - Bob generates a new hash (H1) from the received message.
 - Bob decrypts the digital signature using **Alice's public key** to obtain the original hash (H2).
6. **Comparison**
 - If **H1 = H2**, the signature is valid, proving the sender is authentic and the message has not been modified.
 - If **H1 ≠ H2**, the message has been altered or the signature is invalid.

Advantages of Digital Signature

- **Authentication:** Confirms the identity of the sender.
- **Integrity:** Detects any modification of the message.
- **Non-Repudiation:** Prevents the sender from denying that they sent the message.

Conclusion

Digital signatures provide **authentication, integrity, and non-repudiation** by combining **hash functions** with **asymmetric cryptography**. They are widely used in secure email, online banking, e-commerce, and electronic document signing.

Q21: A college network experiences a ransomware attack due to an employee opening a malicious email attachment. Evaluate the preventive measures that could have minimized the impact. 8 Evaluate (L5)

A ransomware attack occurred because an employee opened a malicious email attachment. The following preventive measures could have significantly minimized the impact of the attack.

1. **Security Awareness Training**
 - Train employees to identify phishing emails, suspicious links, and malicious attachments.
 - Regular awareness programs reduce the chances of human error.
2. **Email Security and Spam Filtering**
 - Use email security gateways to scan and block malicious attachments and phishing emails before they reach users.
 - Enable attachment and URL filtering.
3. **Regular Software Updates**
 - Keep operating systems, applications, and antivirus software updated.
 - Security patches close vulnerabilities that ransomware can exploit.
4. **Antivirus/Endpoint Protection**
 - Install and regularly update antivirus or Endpoint Detection and Response (EDR) solutions.
 - These tools can detect and block ransomware before it executes.
5. **Regular Data Backup**
 - Maintain frequent backups of important data on offline or cloud storage.
 - In case of ransomware, data can be restored without paying the ransom.
6. **Access Control and Least Privilege**
 - Provide users only the permissions required for their work.
 - Limiting user privileges reduces the spread of ransomware across the network.
7. **Network Security Controls**
 - Use firewalls, Intrusion Prevention Systems (IPS), and network segmentation to limit the spread of ransomware between systems.
8. **Overall Evaluation**
 - No single security measure is sufficient. A combination of **employee awareness, email filtering, endpoint protection, regular updates, backups, access control, and network security** provides a strong defense against ransomware and minimizes its impact.

Conclusion

The ransomware attack could have been greatly reduced or even prevented by implementing **multiple layers of security**. Among these, **employee awareness training, secure email filtering, regular backups, updated security software, and strong access controls** are the most effective measures for protecting a college network against ransomware attacks.

Q22: Evaluate the importance of Information Privacy in protecting personal information collected by online service providers. 8

Evaluate (L5)

Information Privacy refers to the protection of personal information collected, stored, processed, and shared by online service providers. It ensures that users have control over how their personal data is used.

Evaluation of Importance

1. **Protects Personal Information**
 - Ensures that sensitive data such as names, addresses, phone numbers, banking details, and passwords are protected from unauthorized access.
2. **Prevents Identity Theft and Fraud**
 - Proper privacy practices reduce the risk of identity theft, financial fraud, and misuse of personal information by cybercriminals.
3. **Builds User Trust**
 - Users are more likely to use online services when they know their personal information is collected and handled responsibly.
4. **Ensures User Consent and Transparency**
 - Service providers should clearly inform users about what data is collected, why it is collected, and obtain user consent before using or sharing it.
5. **Supports Legal and Regulatory Compliance**
 - Organizations must comply with privacy laws and regulations to avoid legal penalties and protect user rights.
6. **Improves Data Security**
 - Privacy practices such as encryption, access control, and secure storage help prevent data breaches and unauthorized disclosure of information.
7. **Enhances Organizational Reputation**
 - Organizations that protect customer privacy maintain a good reputation, increase customer confidence, and encourage long-term relationships.
8. **Overall Evaluation**
 - Information privacy is essential because it protects users' personal data, reduces cyber risks, ensures legal compliance, and strengthens trust between users and online service providers.

Conclusion

Information privacy plays a vital role in protecting personal information collected by online service providers. By implementing **privacy policies, user consent, encryption, access control, and secure data management**, organizations can safeguard user information, comply with legal requirements, and maintain customer trust.

Q23: Evaluate the advantages and disadvantages of Asymmetric Cryptography for secure communication. 8 Evaluate (L5)

Asymmetric Cryptography (Public Key Cryptography) uses **two different keys**:

- **Public Key** – Used for encryption and shared publicly.
- **Private Key** – Used for decryption and kept secret by the owner.

Advantages

1. **Secure Key Distribution**
 - The public key can be shared openly, so there is no need to exchange a secret key securely.
2. **Provides Authentication**
 - Digital signatures verify the identity of the sender and ensure that the message is authentic.
3. **Ensures Confidentiality**
 - Data encrypted with the recipient's public key can only be decrypted using the corresponding private key.
4. **Supports Non-Repudiation**
 - Since only the sender owns the private key, they cannot deny sending a digitally signed message.

Disadvantages

5. **Slower Performance**
 - Asymmetric encryption requires more computational power and is slower than symmetric encryption.
6. **Higher Resource Usage**
 - It consumes more CPU and memory, making it less suitable for encrypting large amounts of data.
7. **Complex Key Management**
 - Public and private keys, along with digital certificates, must be managed securely.
8. **Dependence on Private Key Security**
 - If the private key is lost or compromised, the security of encrypted communication is also compromised.

Overall Evaluation

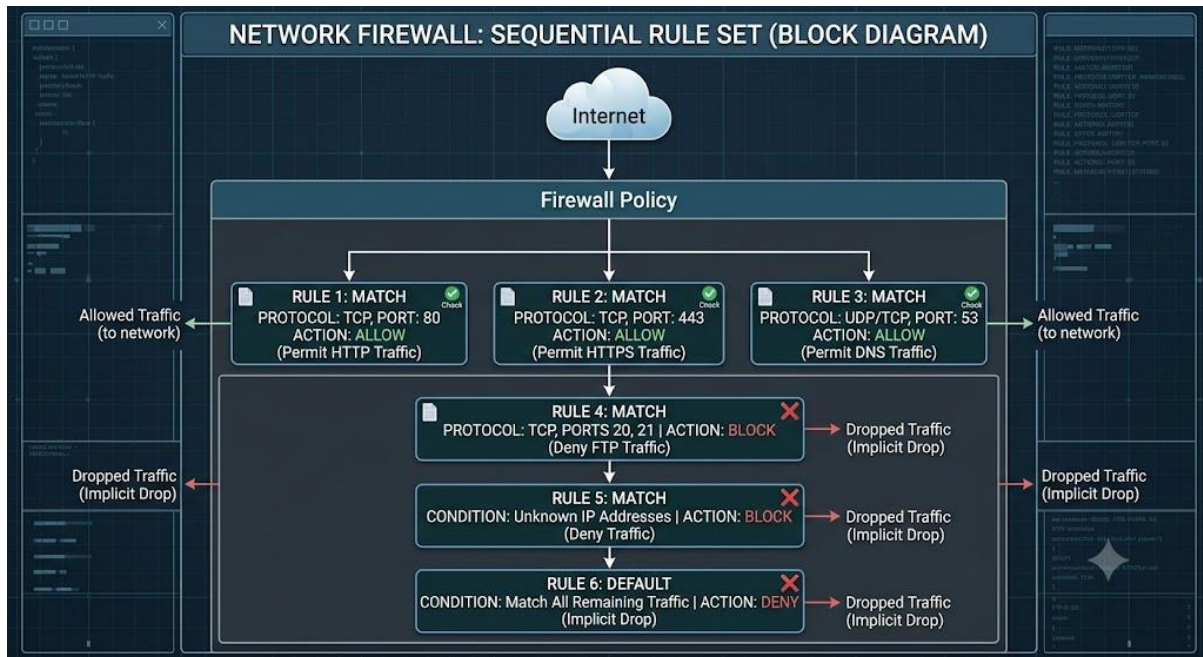
- **Strengths:** Excellent for secure key exchange, authentication, digital signatures, and secure communication over untrusted networks.
- **Limitations:** Slower and more resource-intensive than symmetric cryptography, so it is usually used to exchange encryption keys rather than encrypt large files.

Conclusion

Asymmetric cryptography is highly effective for **secure communication, authentication, confidentiality, and digital signatures**. Although it has higher computational overhead and complex key management, its security benefits make it an essential component of **HTTPS, SSL/TLS, secure email, VPNs, and online banking**.

Q24: Design a firewall rule set for an organization that allows only HTTP, HTTPS and DNS traffic while blocking unauthorized FTP access and unknown IP addresses. 8 Create (L6)

Rule No.	Source IP	Destination	Protocol / Port	Action	Purpose
1	Authorized Internal Network	Web Server	HTTP (TCP Port 80)	Allow	Permit web browsing.
2	Authorized Internal Network	Web Server	HTTPS (TCP Port 443)	Allow	Permit secure web communication.
3	Authorized Internal Network	DNS Server	DNS (UDP/TCP Port 53)	Allow	Allow domain name resolution.
4	Any Source	Any Destination	FTP (TCP Ports 20 & 21)	Deny	Block unauthorized FTP access.
5	Unknown/Unauthorized IP Addresses	Internal Network	Any Protocol	Deny	Prevent access from unknown IP addresses.
6	Any	Any	All Other Traffic	Deny (Default Rule)	Block all traffic not explicitly permitted.



Justification

1. **Allows HTTP (Port 80)** for normal web browsing.
2. **Allows HTTPS (Port 443)** for secure encrypted web communication.
3. **Allows DNS (Port 53)** so users can resolve domain names.
4. **Blocks FTP (Ports 20 & 21)** to prevent unauthorized file transfers.
5. **Blocks unknown or unauthorized IP addresses** to reduce the risk of external attacks.
6. **Uses a Default Deny policy**, ensuring that any traffic not explicitly allowed is automatically blocked.

Conclusion

This firewall rule set follows the **principle of least privilege** by allowing only essential services (**HTTP, HTTPS, and DNS**) while blocking **FTP, unknown IP addresses, and all other unauthorized traffic**, thereby improving the organization's network security.

Q25: Create a Security Policy for an organization covering password policy, Internet usage policy and access control policy. 8 Create (L6)

Security Policy for an Organization

The following security policy helps protect the organization's information systems and data.

1. Password Policy

- Passwords must be **at least 12 characters** long.
- Passwords should include **uppercase letters, lowercase letters, numbers, and special characters**.
- Passwords must not contain personal information such as names or birthdays.
- Passwords should be changed **every 90 days**.
- **Multi-Factor Authentication (MFA)** should be enabled for critical systems.
- Passwords must not be shared with anyone.

2. Internet Usage Policy

- The Internet should be used **only for official and authorized purposes**.
- Users must not visit malicious or unauthorized websites.
- Downloading unauthorized software or files is prohibited.
- Users should not open suspicious email links or attachments.
- Internet activities may be monitored to ensure compliance with organizational policies.

3. Access Control Policy

- Users should be given **only the minimum access permissions** required for their job (**Principle of Least Privilege**).
- Every employee must have a **unique user ID and password**.
- Access to sensitive data should be restricted to authorized personnel only.
- User accounts of employees who leave the organization must be disabled immediately.
- Regular reviews of user accounts and permissions should be conducted.

Benefits of the Security Policy

- Protects confidential organizational data.
- Prevents unauthorized access and cyber attacks.
- Promotes responsible use of organizational resources.
- Improves compliance with security standards and organizational regulations.

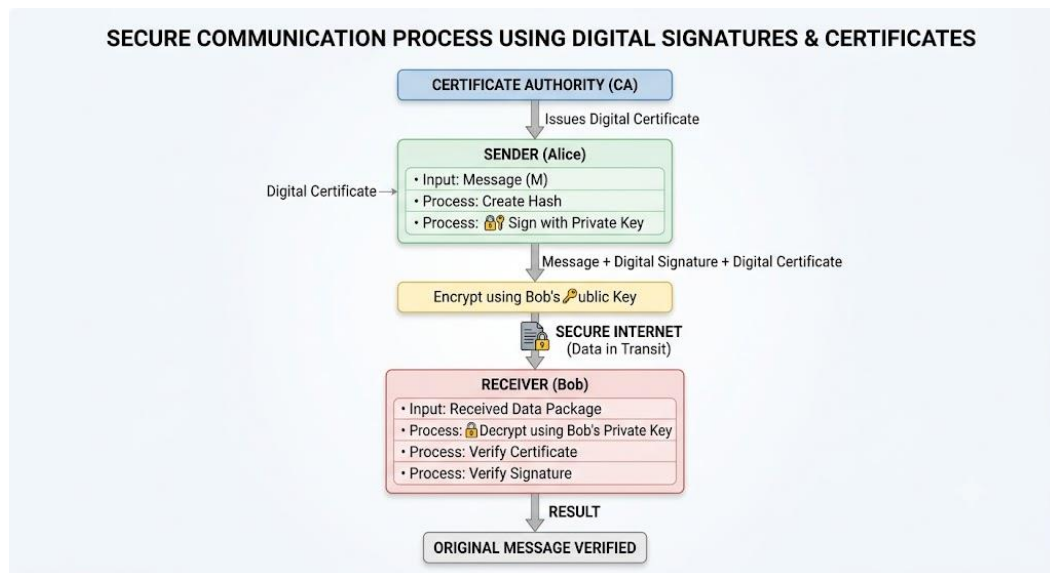
Conclusion

A security policy consisting of a **strong password policy, responsible Internet usage policy, and strict access control policy** helps safeguard organizational information, reduces security risks, and ensures a secure working environment.

Q26: Design a secure communication model using Public Key, Private Key, Digital Signature and Digital Certificate. 8 Create (L6)

Design of a Secure Communication Model

A secure communication model uses **Public Key, Private Key, Digital Signature, and Digital Certificate** to provide **confidentiality, authentication, integrity, and non-repudiation**.



Working of the Model

1. **Digital Certificate Issuance**
 - A **Certificate Authority (CA)** issues a digital certificate to Alice, verifying her identity.
2. **Message Preparation**
 - Alice creates the message and generates its **hash value** using a hash function.
3. **Digital Signature Creation**
 - Alice encrypts the hash using **her private key** to create a **digital signature**.
4. **Message Encryption**
 - Alice attaches the **digital signature** and **digital certificate** to the message.
 - The complete message is encrypted using **Bob's public key**.
5. **Secure Transmission**
 - The encrypted message is sent securely over the Internet.
6. **Message Decryption**
 - Bob decrypts the message using **his private key**.
7. **Certificate Verification**
 - Bob verifies Alice's **digital certificate** using the Certificate Authority (CA) to confirm Alice's identity.
8. **Digital Signature Verification**
 - Bob decrypts the digital signature using **Alice's public key** and compares the received hash with a newly generated hash.
 - If both hashes match, the message is authentic and has not been modified.

Security Services Provided

Component	Purpose
Public Key	Encrypts data for secure transmission.
Private Key	Decrypts data and creates digital signatures.
Digital Signature	Ensures authentication, integrity, and non-repudiation.
Digital Certificate	Verifies the identity of the sender through a trusted Certificate Authority (CA).

Conclusion

This secure communication model combines public key encryption, private key decryption, digital signatures, and digital certificates to provide confidentiality, authentication, integrity, and non-repudiation, making it suitable for online banking, secure email, e-commerce, and digital document exchange.

Q27: Create a comprehensive Internet Security plan for an educational institution using firewall protection, antivirus software, software updates, MFA and backup mechanisms. 8 Create (L6)

Comprehensive Internet Security Plan for an Educational Institution

A comprehensive Internet Security plan protects the institution's computers, servers, networks, and student data from cyber threats.

1. Firewall Protection

- Install **hardware firewalls** at the network gateway to protect the entire campus network.
- Use **software firewalls** on individual computers and servers.
- Allow only authorized services (HTTP, HTTPS, DNS) and block unauthorized ports and suspicious traffic.

2. Antivirus and Anti-Malware Protection

- Install licensed antivirus software on all computers and servers.
- Enable real-time protection and automatic virus definition updates.
- Perform regular system scans to detect and remove malware.

3. Regular Software Updates

- Keep operating systems, applications, browsers, and antivirus software up to date.
- Apply security patches immediately to fix known vulnerabilities.
- Enable automatic updates whenever possible.

4. Multi-Factor Authentication (MFA)

- Enable MFA for faculty, staff, and administrator accounts.
- Require a second authentication factor such as an OTP or authentication app in addition to the password.
- Use MFA for access to email, learning management systems, and administrative portals.

5. Data Backup Mechanism

- Perform regular backups of student records, examination data, and institutional documents.
- Store backups on secure offline storage or trusted cloud storage.
- Periodically test backup restoration to ensure data can be recovered successfully after a failure or ransomware attack.

6. Security Awareness and Access Control

- Conduct cybersecurity awareness programs for students and staff.
- Enforce strong password policies and the **Principle of Least Privilege**.
- Restrict access to sensitive systems based on user roles.

7. Network Monitoring

- Monitor firewall logs, system logs, and network traffic for suspicious activities.
- Use Intrusion Detection/Prevention Systems (IDS/IPS) to detect and block attacks.

8. Incident Response and Recovery

- Establish an incident response team to handle security incidents.
- Isolate infected systems, remove threats, restore data from backups, and review the incident to prevent recurrence.

Summary Table

Security Measure	Purpose
Firewall Protection	Blocks unauthorized network access and malicious traffic.
Antivirus Software	Detects and removes malware and ransomware.
Software Updates	Fixes security vulnerabilities and improves protection.
Multi-Factor Authentication (MFA)	Prevents unauthorized account access.
Backup Mechanism	Ensures data recovery after failures or cyber attacks.

Conclusion

A comprehensive Internet Security plan combining firewall protection, antivirus software, regular software updates, Multi-Factor Authentication (MFA), and reliable backup mechanisms provides strong protection against cyber threats. Along with user awareness, access control, and continuous monitoring, it ensures the confidentiality, integrity, and availability of the educational institution's information systems and data.